

## Week5 Session-4 Interview Questions

### 1. What is the difference between synchronous and asynchronous programming?

Difference between synchronous and asynchronous programming:

- Synchronous programming: Operations are executed one at a time. The program waits for a task to complete before moving to the next one, which can block the main thread.
- Asynchronous programming: Operations are executed without blocking the main thread. Tasks can run concurrently, improving performance and responsiveness.

### 2. Why is asynchronous programming important in Node.js?

Asynchronous programming allows Node.js to handle multiple I/O operations (like reading files or accessing a database) efficiently without blocking the main thread. This is crucial for building high-performance, scalable applications.

### 3. What is a callback in Node.js?

A callback is a function passed as an argument to another function, which gets executed after the completion of an asynchronous operation.

Example:

```
const fs = require('fs');
fs.readFile('file.txt', 'utf8', (err, data) => {
  if (err) throw err; console.log(data);
});
```

### 4. How does Node.js handle asynchronous tasks using callbacks?

Node.js uses an event loop to manage asynchronous tasks. When an operation (e.g., file reading) completes, the callback function is pushed to the event queue and executed when the call stack is empty.

### 5. What is callback hell?

Callback hell occurs when multiple nested callbacks are used, making the code difficult to read, maintain, and debug.

Example:

```
fs.readFile('file1.txt', 'utf8', (err, data1) => {
  if (err) throw err;
  fs.readFile('file2.txt', 'utf8', (err, data2) => {
    if (err) throw err;
    fs.writeFile('result.txt', data1 + data2, (err) => {
      if (err) throw err;
      console.log('Files combined!');
    });
  });
});
```

### 6. How can you avoid callback hell in Node.js?

To avoid callback hell:

- Use **promises** or the Promise API.
- Use **async/await** for cleaner syntax.
- Modularize your code into smaller functions.
- Use libraries like **async.js**.

### **7. What is a promise in Node.js, and how does it solve callback hell?**

A promise is an object that represents the eventual result (resolved or rejected) of an asynchronous operation. Promises help flatten nested callbacks and improve readability.

### **8. How does the event loop in Node.js work?**

The event loop is a mechanism that handles asynchronous operations in Node.js. It continuously checks the event queue and moves callbacks into the call stack for execution when the stack is empty.

### **9. Why is error handling crucial in asynchronous programming?**

Asynchronous operations do not throw errors directly to the main thread. Proper error handling ensures that the application does not crash unexpectedly and allows graceful error recovery.